

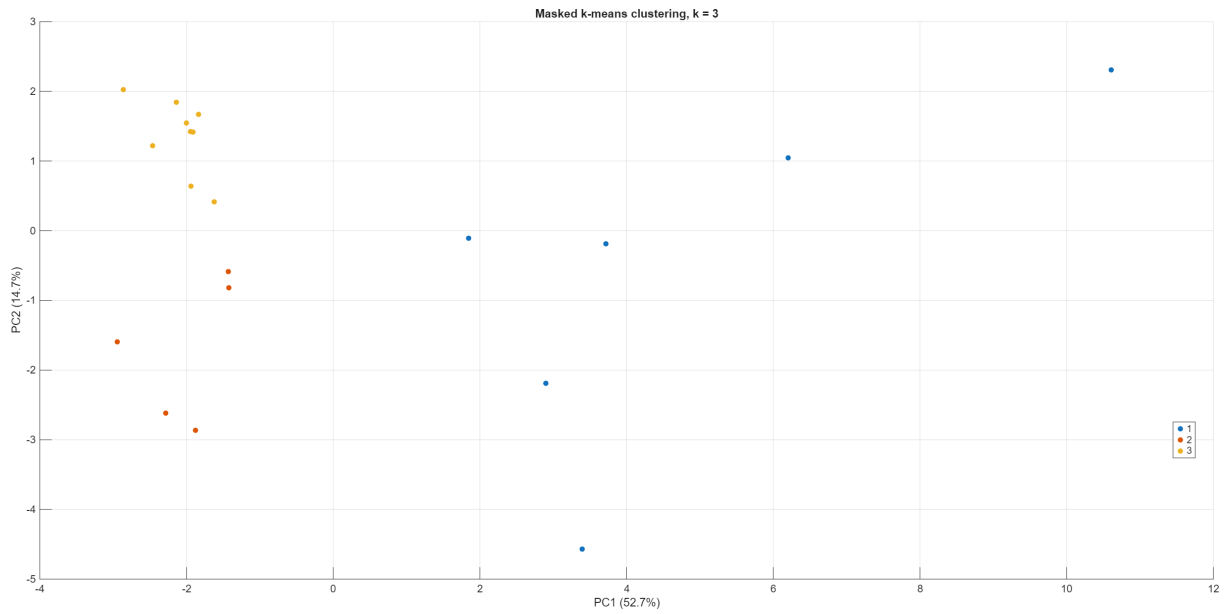
ERG Signal Analysis via MATLAB Machine Thinking

Version 0.2 – Work in Progress

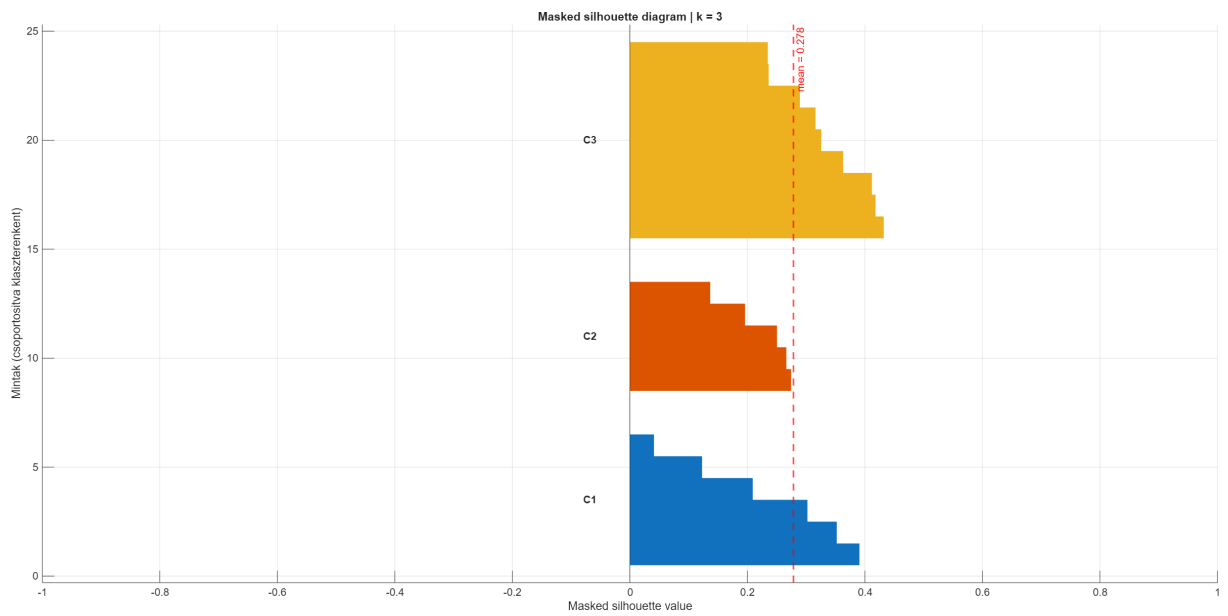
Levente Csaki

2026. április 19.

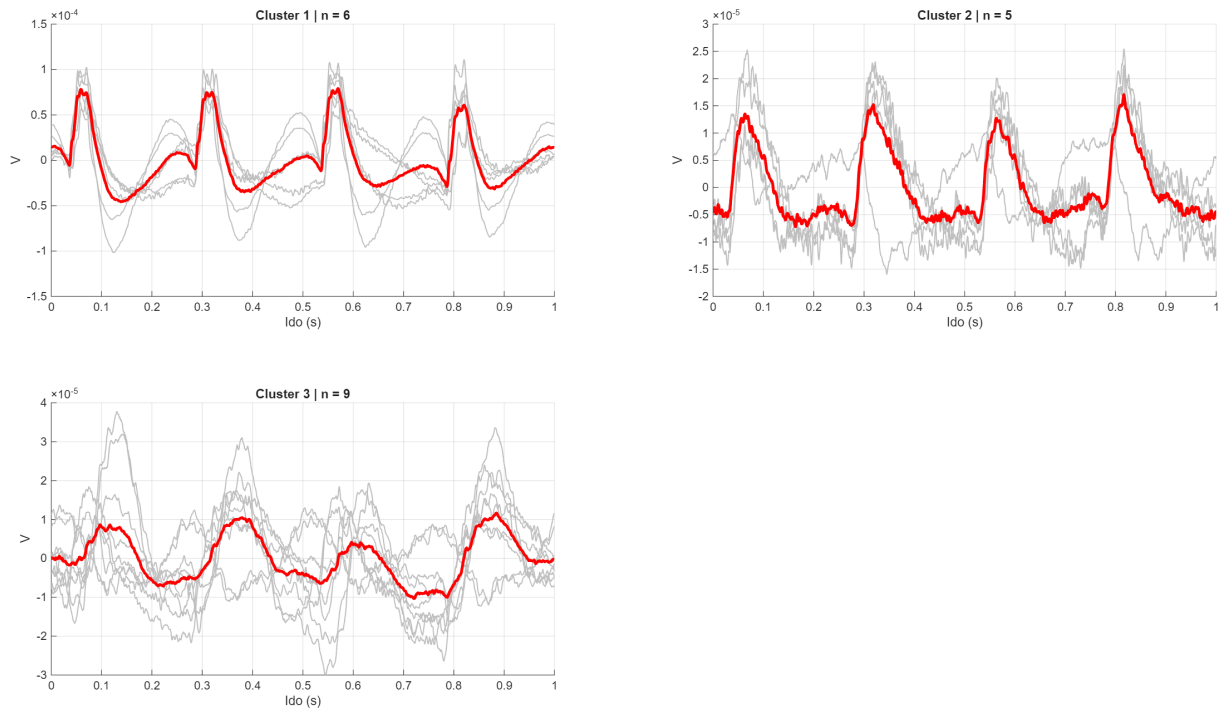
1. $K=3$



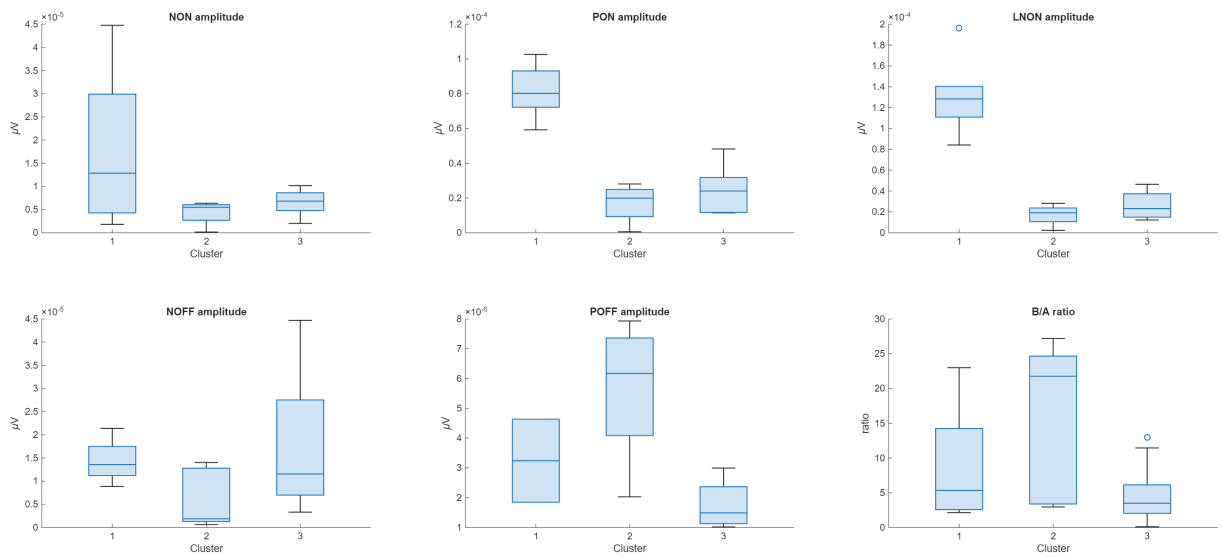
1. ábra. PCA at $k = 3$



2. ábra. Silhouette scores at $k = 3$, mean avg. 0.278 (low)



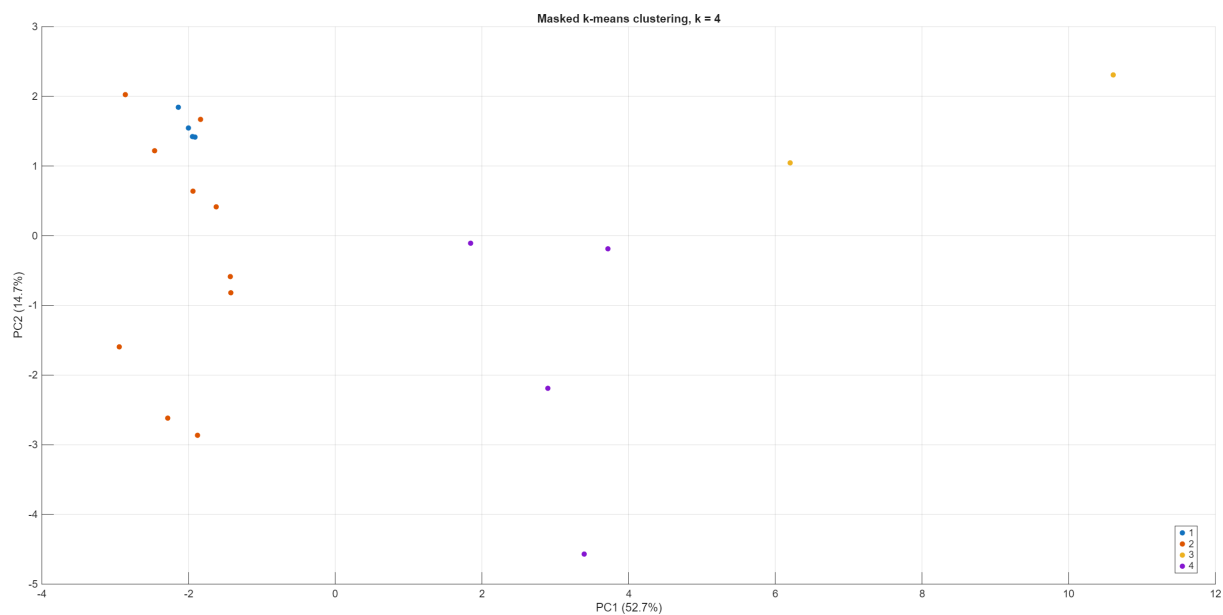
3. ábra. Averaged graphs in each cluster



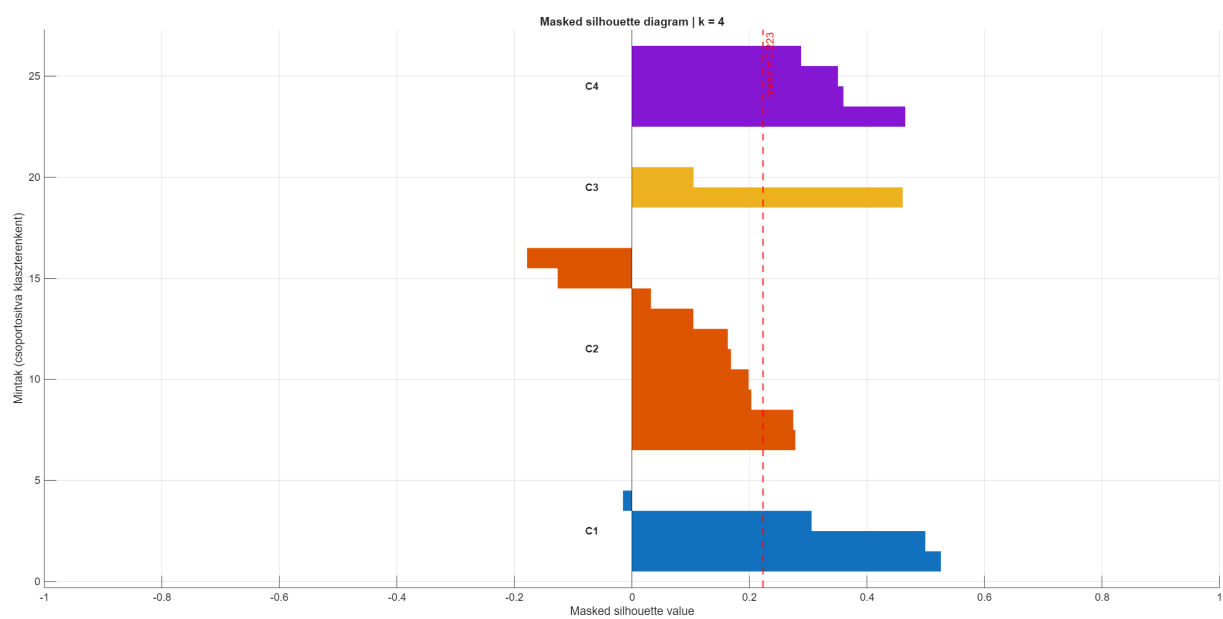
4. ábra. Some properties of clusters for comparison and evaluation

This cluster number (3) produces the clearest results, even though SS is low.

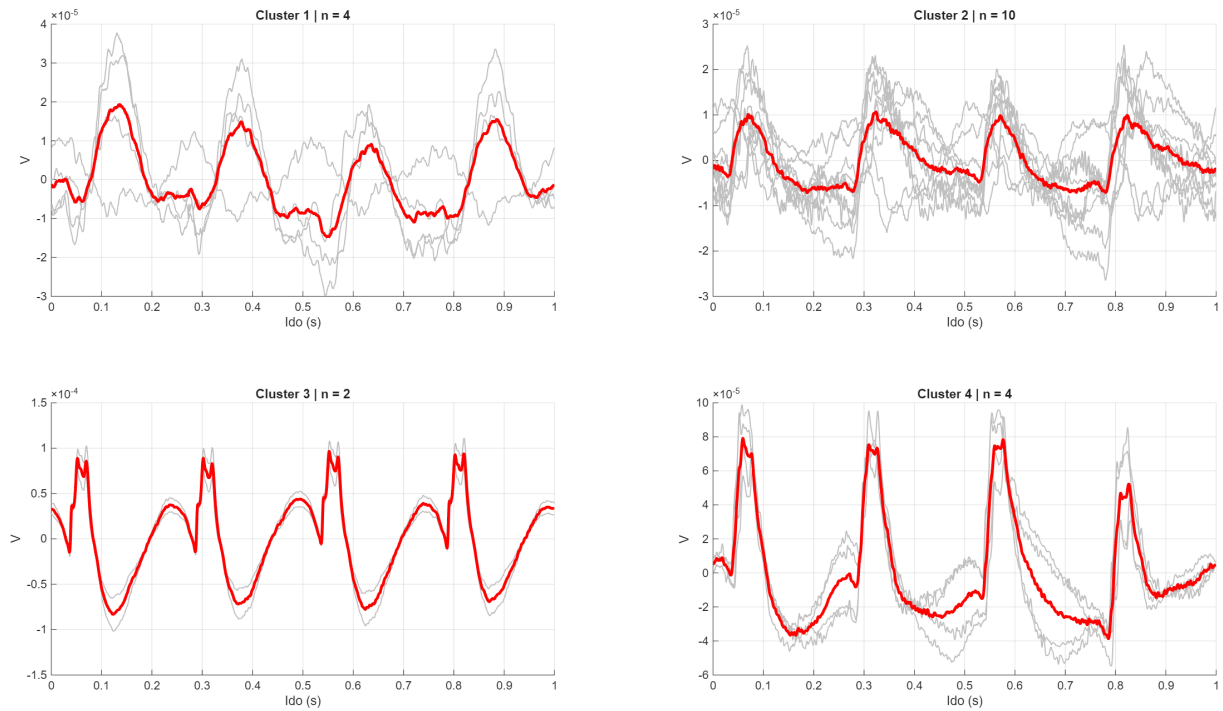
2. K=4



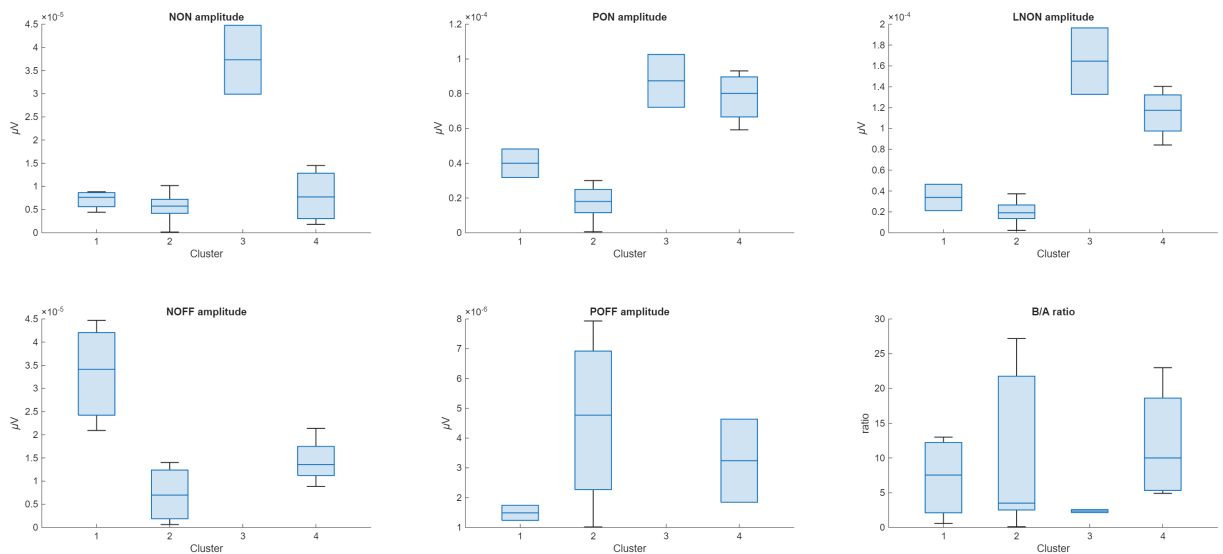
5. ábra. PCA at $k = 4$, anomaly at C1 and C2



6. ábra. Silhouette scores at $k = 4$, mean avg. 0.223 (even lower, big assimetry in C2)



7. ábra. Averaged graphs in each cluster



8. ábra. Some properties of clusters for comparison and evaluation

Improvement: some waveforms got separated as the number of clusters increased. Drawback: other previously separately grouped elements got grouped together.

Full code and data pipeline available upon request. Repository: [GitHub repository](#)